Jaggy *et al.*
International Journal of STEM Education (2025) 12:53
https://doi.org/10.1186/s40594-025-00576-w

International Journal of STEM Education

RESEARCH

Open Access

Teaching quality in STEM education: differences between in- and out-of-school contexts from the perspective of gifted students

Ann-Kathrin Jaggy[1*], Wolfgang Wagner[1], Tim Fütterer[1], Richard Göllner[1,2] and Ulrich Trautwein[1]

Abstract

Background Teaching quality plays a critical role in effective STEM education, and improving teaching quality is often considered the most promising approach for boosting STEM achievement. However, teachers frequently encounter substantial heterogeneity in students' aptitudes, interests, and needs, thus rendering it challenging to maintain high-quality instruction for all learners. Consequently, it has been argued that additional out-of-school programs that cater to a more homogeneous student group might better meet students' needs, resulting in higher teaching quality. At the same time, by attending high-quality out-of-school programs, students' evaluations of their regular school lessons might be negatively affected via comparison or reference effects. However, there is a lack of studies on teaching quality in regular classes versus out-of-school programs.

Method The present study used data from a specialized out-of-school STEM enrichment program for highly gifted students. We compared the teaching quality in regular school classes and the enrichment program as reported by a total of 67 gifted students (IQ $M = 133.7$; $SD = 7.2$) with the perceived teaching quality in regular classes reported by their peers ($N = 268$) who were nominated for the program but were not enrolled because they scored lower on a cognitive ability test. More specifically, we compared the children's perceptions of the motivational climate of both contexts and the teaching quality in students' regular lessons (i.e., math classes) and the STEM enrichment program.

Results Results showed that students participating in the out-of-school STEM enrichment program reported higher motivational climate and teaching quality in the enrichment program than in their regular (math) classes. At the same time, the perceptions of the motivational climate and teaching quality in the regular school setting of students who attended the enrichment program were similar to the perceptions of their peers who did not have access to the specialized enrichment program.

Conclusion The findings indicate that the out-of-school STEM enrichment program we examined constitutes a valuable learning environment to meet gifted students' needs and enhance their learning experiences. We discuss the insights we gained from gifted education in the context of different educational settings and contribute to a better understanding of perceived differences in teaching quality.

Keywords STEM education, Teaching quality, Giftedness, Secondary school

Introduction

Teaching quality plays a critical role in effective science, technology, engineering, and mathematics (STEM) education, and improving teaching quality is often considered the most promising approach for boosting STEM students' positive development and learning outcomes

*Correspondence:
Ann-Kathrin Jaggy
ann-kathrin.jaggy@uni-tuebingen.de
[1] University of Tübingen, Tübingen, Germany
[2] University of Potsdam, Potsdam, Germany

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

(Hamre et al., 2013; Huang et al., 2022; Wagner et al., 2016). Thus, it is important to provide students with high-quality teaching.

However, teachers frequently encounter substantial heterogeneity in students' aptitudes, interests, and needs, which makes it difficult to maintain high-quality instruction for all learners (Suprayogi et al., 2017). Gifted students, for instance, have special educational needs that are not always met in regular classes (Gomez-Arizaga et al., 2020). Consequently, it has been argued that gifted students may need enriched environments to realize their full potential (Kultusministerkonferenz [KMK], 2016; Reis & Renzulli, 2023).

Out-of-school programs catering to a more homogeneous student group are believed to offer a potential solution to better meet gifted students' needs (Reis & Renzulli, 2023; Steenbergen-Hu et al., 2016). Increasing evidence suggests that such programs have positive effects on several learning outcomes (e.g., Trautwein et al., 2023). A recent systematic review by Neher-Asylbekov and Wagner (2023), for example, highlights the positive impact of hands-on activities and high-quality instruction in out-of-school STEM environments on student interest, supporting the notion that these settings are particularly effective for fostering motivation. Consequently, it is also plausible that such learning environments would result in positive evaluations by the participating students regarding various aspects of these programs. However, when it comes to perceived teaching quality in such programs, there is a dearth of empirical studies. In this study, we used the internationally prominent *Three Basic Dimensions* (TBD) (of teaching quality) framework (Klieme et al., 2009; Praetorius et al., 2018) to systematically investigate perceived teaching quality across diverse learning environments. We investigated students' perceived teaching quality of a specialized out-of-school STEM enrichment program for gifted secondary students: the Hector Seminar (Heller, 2009). We contrasted the perceived teaching quality of the extracurricular specialized STEM enrichment program with that of the regular school setting. In addition, we investigated differences in the perceived teaching quality of regular schools between students who attended the specialized enrichment program and their peers who did not participate in the enrichment program.

Teaching quality in STEM education

Teaching quality is crucial for successful student learning processes and learning outcomes, such as achievement, motivation, and well-being (e.g., Wagner et al., 2016), and can be conceptualized through theoretically founded and empirically well-tested core dimensions (Christ et al., 2024). In this article, we followed the TBD framework in which teaching quality is represented by the core dimensions of *effective classroom management*, *cognitive activation* of students, and *student support* (Christ et al., 2024; Klieme et al., 2009; Praetorius et al., 2018, 2020a, 2020b; Senden et al., 2022). These three core dimensions are considered valid across academic subjects (e.g., STEM subjects and language subjects), although there are subject-specific distinctions (e.g., how students are cognitively engaged; Praetorius & Gräsel, 2021; Praetorius et al., 2020a, 2020b).

Effective classroom management

Effective classroom management establishes a smooth learning process by ensuring that learners understand the goals and structure of a lesson, preventing disruptions or conflicts between students as much as possible, and involving as many learners as possible (Korpershoek et al., 2016; Kunter et al., 2007; Marder et al., 2023). In effective classroom management, rules and procedures are explicitly established, and lessons follow well-planned structures, making disruptions and interruptions less likely. The high level of structure, clarity of expectations, and reliability make it easier for learners to develop a sense of control and actively participate in the lesson (Hamre et al., 2013).

Cognitive activation

Cognitive activation is pronounced when teaching is characterized by practices that encourage students to engage in high-level thinking and understanding. Such practices involve presenting challenging tasks, encouraging deep reasoning, promoting active problem-solving, engaging intensively with the learning material, and connecting the material to students' prior knowledge and their understanding of essential learning content and concepts (Baumert et al., 2010; Fütterer et al., 2023; Holzberger & Prestele, 2021). New knowledge can be acquired or existing misconceptions can be corrected during learning activities. Such cognitively activating teaching practices often involve appropriate learning strategies and metacognitive regulation of learning activities.

Student support

Student support is expressed when teachers provide appropriate help with comprehension problems and when interactions between teachers and learners are characterized by respect and appreciation (e.g., Pianta & Hamre, 2009; Praetorius et al., 2018). Particularly effective components of student support include the students' perceptions that the teacher is interested in their positive development, the individual feedback that comes with such interest, and the use of adaptively designed aids to help individual learners achieve challenging learning

goals. A high level of student support enables children and adolescents to expand their motivational resources, experience themselves as at least partially autonomous actors, and acquire a positive self-image.

Given the focus of this study on specialized out-of-school enrichment programs, two specific subdimensions of student support, *adaptivity* and *interestingness*, may be particularly important. Adaptivity describes how teachers align the learning environment with students' abilities, learning preferences, prior knowledge, and interests (Corno, 2008; Dumont, 2019; Fütterer et al., 2024). Interestingness of teaching refers to how interesting the arrangement of a task or instruction is in general (Patall, 2013) and is an essential characteristic of specialized out-of-school programs that are explicitly designed to spark, maintain, and foster students' interests.

Interindividual differences in perceived teaching quality

According to the offer-and-use model of learning (Helmke, 2007), teaching and learning is an inherently interactive process in which teachers provide learning opportunities that require students' active engagement to be effective (Praetorius et al., 2020a, 2020b). Thus, teaching quality is not just the teacher's responsibility, but also involves the students' active participation.

This engagement with learning opportunities, in turn, depends strongly on individual student characteristics, such as cognitive abilities, motivation, and achievement levels (Göllner et al., 2020). For example, highly interested students might invest more active learning time in learning material that is challenging and may perceive an activity as more cognitively activating than less interested students, indicating systematic variation that depends on student characteristics. The same instructional setting can thereby be perceived differently by individual students (Göllner et al., 2018).

Correspondingly, previous studies have revealed relationships between classroom composition and teaching quality. Studies have found that class average achievement and motivation are positively associated with perceived teaching quality (e.g., Arens & Möller, 2016; Bardach et al., 2019; Fauth et al., 2020; Göllner et al., 2020; Hachfeld & Lazarides, 2021). For example, in a study by Hachfeld and Lazarides (2021), the average level of student achievement in class was positively related to student-perceived cognitive activation ($\beta = 0.46$) but not with effective classroom management or student support.

At the same time, class heterogeneity in cognitive abilities or achievement has been shown to be negatively associated with teaching quality (Bardach et al., 2019; Decristan et al., 2017), supporting the assumption that teaching might not be adaptive enough (Röhl et al., 2024). Decristan et al. (2017), for instance, showed that greater class heterogeneity in students' reading comprehension was associated with lower levels of perceived cognitive activation ($r = -.41$) but not with instructional support or effective classroom management. Moreover, Arens and Möller (2016) found that instructional quality was strongly related to students' individual achievement. This variation highlights the challenge for educators: how to design instruction that meets the diverse needs of all learners (OECD, 2016), including gifted children (Kultusministerkonferenz [KMK], 2016). When teachers follow a *one-size-fits-all* approach, for example, by providing direct instruction that targets the group's average level of competence, students at both ends of the ability distribution are often disadvantaged (Tieso, 2003; Tomlinson et al., 2003). High-achieving students may quickly lose interest and become disengaged, whereas low-achieving students may struggle to keep up, feeling overwhelmed and frustrated.

To address these disparities, teachers should ensure that all students are equitably challenged relative to their abilities. To this end, tailored instruction via differentiation and adaptive teaching is essential (Tomlinson et al., 2003). Tailoring instruction can occur on various levels: microlevel adaptation, which involves real-time decisions made during classroom teaching such as providing different tasks; and macrolevel adaptation, which refers to comprehensive, structured programs designed to support groups of students with similar needs, such as gifted learners, with the aim of promoting individual growth and reducing educational disadvantages (Hardy et al., 2019; Steenbergen-Hu et al., 2016).

Specialized out-of-school enrichment programs for gifted students

Informed by current theoretical talent development models, such as the Talent Development in Achievement Domains (TAD) framework (Preckel et al., 2020), this study conceptualizes giftedness as high domain-specific (cognitive) potential that can develop into high achievement. In the STEM domain, this potential typically manifests through advanced abstract reasoning, rapid knowledge acquisition, and strong problem-solving capabilities. According to the TAD framework, the realization of this potential into high achievement depends on the dynamic interplay between stable individual traits (e.g., motivation, self-concept), specific aptitudes, and contextual influences, such as instructional quality and learning opportunities, including specialized out-of-school enrichment programs.

Specialized out-of-school enrichment programs for gifted students are designed to enhance and deepen learning by going beyond the standard curriculum to provide opportunities that address gifted learners' unique

aptitudes, interests, and needs (Reis & Renzulli, 2023; Trautwein et al., 2023). The aim is to provide broader (horizontal enrichment) or more advanced (vertical enrichment) learning, by adapting the teaching methods, materials, and pace to the group's specific needs (Reis & Renzulli, 2023).

Additional STEM enrichment programs are often characterized by inquiry-based learning and project-based learning. Inquiry-based learning is a teaching method in which students work in a manner similar to the way scientists work (Blanchard et al., 2010). Project-based learning (Sahin, 2013) provides gifted students with opportunities for individualized pacing and comprehensive, interdisciplinary exploration of knowledge with tangible, meaningful applications.

Another characteristic of specialized out-of-school enrichment programs is their formation of student groups that are relatively homogenous regarding cognitive abilities and specific interests (Tieso, 2003). This homogeneity can enable educators to tailor their teaching more effectively (e.g., Harker & Tymms, 2004), potentially fostering a better fit between offered learning opportunities and student engagement (cf. Weinert & Helmke, 1995).

At the same time, the unique composition of such groups can also have a direct impact on the effectiveness of instructional strategies and the learning environment as a whole. Students actively contribute to the quality of teaching by engaging in the learning process, providing feedback, and collaborating with their peers (Weinert & Helmke, 1995). Moreover, in high-ability groups, teachers can devote more time to instruction (i.e., student support or cognitive activation), spend less time on management and discipline (i.e., effective classroom management). This reallocation of time can, in turn, allow teachers to cover more of or go beyond the curriculum at a faster pace (Tieso, 2003).

Against this background, one might assume that these specialized out-of-school programs for gifted students are characterized by high-quality teaching. However, to the best of our knowledge, the teaching quality of such programs has yet to be empirically evaluated. So far, empirical studies have focused on the effects of such specialized enrichment programs or ability grouping for gifted students on students ' development, with meta-analytic evidence of substantial positive effects on academic achievement and socioemotional development (Kim, 2016; Kulik & Kulik, 1982; Steenbergen-Hu et al., 2016). Complementing this, other research has shed light on gifted students' perspectives regarding effective instruction. Qualitative studies have highlighted that gifted students particularly value teacher expertise, challenging content, and positive student–teacher relationships (e.g., Khalil & Accariya, 2016). Furthermore, initial descriptive studies have begun to assess gifted students' views on specific classroom practices such as differentiation, the provision of challenging activities, and the overall classroom climate (e.g., Kahveci & Akgül, 2014).

Reference effects: comparisons of learning environments

As demonstrated, specialized enrichment programs can be expected to facilitate instruction and thus might provide a high(er) teaching quality (evaluation). However, experiencing additional learning environments with potentially higher teaching quality might influence students' evaluations of their regular classes, as subjective evaluations are shaped by comparative and contextual factors, such as reference effects (Marsh, 1987; Möller, 2024).

Generally, students experience various teachers and teaching styles across different subjects, and they might compare these different teachers with each other (Göllner et al., 2018). In our case, students might use their experience with the specialized enrichment program as a reference for an external comparison and evaluation of their perceived teaching quality in their regular classes.

On the one hand, this might lead to decreases in perceived teaching quality of the regular classes. For example, if students perceive the teaching environment in this highly competitive environment as more effective, it might negatively influence their evaluations of the quality of their regular classes (Lichtenstein & Slovic, 2006). In addition, children attending specialized enrichment programs may feel personally connected to the program, leading to ingroup–outgroup comparisons (Tajfel & Turner, 1979), which, in turn, may lead to lower ratings of teaching quality in their regular classes. Such an outcome would be an unfavorable side effect of attending the specialized enrichment program that should be addressed and should thereby be investigated.

On the other hand, attending a highly competitive environment might lead to increases in perceived teaching quality of the regular classes, for example, via decreases in students' self-concepts due to the Big-fish-little-pond effect (BFLPE; Marsh et al., 2008). The BFLPE states that students evaluate their abilities in relation to their peers, leading to lower self-concepts in more competitive environments, despite students' actual competence. Transferred to teaching quality evaluations, students with lower self-concepts might feel more competent in their regular school, engage more in their regular classes, and thus perceive the teaching quality in school as higher than before (Weinert & Helmke, 1995).

Previous studies have shown that well-known and established comparison processes, such as the dimensional comparison effect (Möller, 2024), indeed generalize to and influence perceived teaching quality. For example, it has been shown that students' grades and their teaching-quality ratings in mathematics and German language classes show positive within-subject associations, e.g., math grade was positively associated with students' math teaching quality, and negative between-subjects associations, e.g., math grades were negatively associated with student ratings of German teaching quality (Arens & Möller, 2016; Jaekel et al., 2020). These studies demonstrate that students' evaluations of instructional quality relate not only to their achievement within that subject, but also systematically to their achievement in other academic domains.

Taken together, attending a high-quality enrichment program could trigger various comparison effects that shape how students evaluate their regular school experiences. However, such comparison and reference effects specifically between extracurricular enrichment and regular school contexts remain empirically unexplored. As a first step toward a better understanding of the potential side-effects of attending extracurricular learning programs, we investigated whether students attending versus not attending a specialized enrichment program perceived the regular school context differently.

The present study

This study was designed to fill a gap in the literature by comparing perceived teaching quality between the regular STEM school setting and a specialized STEM enrichment program designed for gifted students: the Hector Seminar. Focusing on in- and out-of-school teaching quality, this study ultimately contributes to a better understanding of how (gifted) students perceive teaching quality in diverse STEM learning environments. On the basis of the literature discussed above, we addressed two main research questions related to perceived teaching quality in the three basic dimensions introduced above: adaptivity, interestingness of lectures, and the motivational climate of the specialized program.

RQ1: Do gifted students give higher evaluations of the teaching quality of a specialized extracurricular STEM enrichment program (i.e., the Hector Seminar) than their regular school setting?

We specifically hypothesized that gifted students attending the Hector Seminar (i.e., Hectorians) would perceive the six measured indicators of teaching quality (i.e., effective classroom management, cognitive activation, student support, adaptivity, interestingness, and motivational climate) in the out-of-school specialized program higher than in their regular school classes (*Hypotheses 1a-f*).

RQ2: Is there a difference in perceived teaching quality in regular classrooms between gifted students who attended the Hector Seminar and students who did not attend the Hector Seminar?

The second research question was exploratory, as several outcomes were plausible: Hectorians might perceive the six measured indicators of teaching quality in regular classrooms lower due to an unfavorable comparison (i.e., a reference effect; Marsh et al., 2008) with the teaching quality experienced in the Hector Seminar. Alternatively, Hectorians might give higher ratings of regular classroom teaching quality if they are generally more motivated by attending the seminar or the BFLPE. Finally, it is also possible for these opposing effects to balance each other out, resulting in no significant difference between Hectorians and non-Hectorians.

Method

The data originated from an ongoing large-scale study on talent development in secondary school (Jaggy et al., 2025). Data were collected from talented academic track students in the German federal state of Baden-Württemberg. We focused on cross-sectional data from the first two study cohorts (sixth and seventh graders).

Transparency and openness

We adhere to the Journal Article Reporting Standards (JARS; Kazak, 2018). All data, analysis codes, and research materials are available at the Open Science Framework: https://osf.io/4y6qp/files. Data were analyzed with *R* (version 4.4.1; R Core Team, 2024). The current study was not preregistered. The Ethics and Data Protection Commission of the University of Tübingen approved the study (A2.5.4-239.3_hb). All participants provided written informed consent after being fully informed about the study procedures and their right to withdraw. ChatGPT (OpenAI, 2024) was used to edit for grammar and style but not for text generation or research.

Participants and procedure

Participants were recruited from a two-step selection procedure (nomination and testing) for an enrichment program for gifted secondary school students (i.e., the Hector Seminar), located in a smaller area of the German federal state of Baden-Württemberg. In Step 1, using a checklist, STEM teachers from 80 cooperating schools were asked to nominate the best 10% of fifth-grade students on the basis of perceived STEM competencies, intellect, creativity, and social skills. In Step 2 of the selection process, these students were tested for

cognitive potential (in STEM), using a giftedness test battery. All of the students nominated in the first step from two cohorts (2022 and 2023) were invited to participate in the current study. In total, $N=357$ sixth and seventh graders participated in the cross-sectional online assessment of teaching quality, which is the main database of the current study. Due to missing values in all teaching quality scales, $n=13$ participants were excluded from the analysis. Another $n=9$ students had to be excluded, as they were wrongly classified as non-Hectorians during data collection and thus received the wrong questionnaires. The final sample included $N=335$ participants (41.6% girls, $M_{age}=12.17$ years, $SD=0.53$). Table 1 provides an overview of the sample characteristics across cohorts.

The overall assessment consisted of a cognitive ability test battery and an online questionnaire conducted at different measurement points for the two cohorts. Students' cognitive abilities were assessed with standardized tests at the beginning of sixth grade in both cohorts (Cohort 1: September 2022; Cohort 2: September 2023; see Fig. 1). Perceived teaching quality was assessed with an online questionnaire, administered in September 2023 for Cohort 1 (seventh grade) and January 2024 for Cohort 2 (sixth grade).

The Hector Seminar

The Hector Seminar is a STEM enrichment program for academic track students (i.e., students enrolled in the university track in high school). Financed by the Hector Foundation II, after which it is named, and supported by the Regional Council of Karlsruhe, the program is offered free of charge to participating families. It focuses on the long-term and holistic promotion of STEM-related development through vertical and horizontal enrichment from Grade 6 onwards until graduation. The program operates across four designated hosting school locations within the nomination area. Once students are enrolled in the program, they meet weekly in one of four groups of 20 students for 2 h of STEM enrichment, each led by a tandem of two STEM teachers in team-teaching. The team-teaching collaboration involves the teachers jointly planning course content, co-delivering lessons, and engaging in shared post-session debriefing and reflection.

The approx. 25 instructors of the Hector Seminar are STEM teachers holding a university degree in teacher education. They are regularly employed at academic track schools and are seconded by the Regional Council for half of their teaching load to the program, while most continue their regular teaching duties at their schools with the remaining half of their deputation. While formal advanced certification in gifted education is not a mandatory selection criterion for Hector Seminar instructors, the program prioritizes a high level of established professional expertise in STEM (teaching). For most of the current seminar leaders, this foundational expertise is often complemented by substantial, multi-year practical experience with gifted students within the Hector Seminar. Consequently, while their core pedagogical and subject-matter proficiency aligns with that of other academic track teachers, it is distinctly enhanced by their specialized experience within the Hector Seminar's curriculum and with its gifted student population.

The Hector Seminar integrates various didactic elements, combining challenging and gender-fair course content with a strong focus on interdisciplinary approaches that meaningfully integrate all STEM disciplines and require students to unite knowledge from multiple domains (see also Heller, 2009, p. 140). The core components of the seminar are inquiry-based learning and project-based learning (Blanchard et al., 2010; Sahin,

Table 1 Description of the sample across cohorts

Group	*N*	Girls (%)	Age (in years) *M* (*SD*)	Hectorians *n*
Cohort 1	162	46.9	12.56 (0.37)	29
Cohort 2	173	34.7	11.78 (0.36)	38

N number of participating students, *Hectorians* number of students attending the Hector Seminar

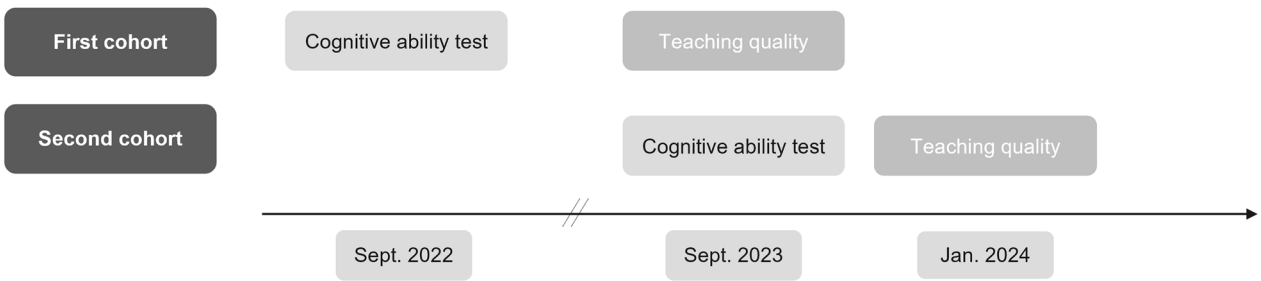


Fig. 1 Study design. The first cohort refers to students who evaluated the teaching quality in Grade 7 and the second cohort refers to students who evaluated the teaching quality in Grade 6

2013). The integration of all four STEM disciplines is centrally achieved through the deliberate selection and implementation of interdisciplinary projects that tackle complex STEM tasks, rather than addressing them as isolated subjects. The Hector Seminar project *boolean algebra*, for example, is initially pure mathematics, but with applications in circuit algebra. There are therefore connections to physics, in particular to electronics, but also to technology and to computer science (see Heller, 2009, p. 82). The course pedagogy emphasizes the promotion of motivation, cognitive development, and self-regulation, as well as the opportunity to gain practical experience (e.g., through mentoring and job shadowing).

The Hector Seminar's capacity to consistently implement these ambitious pedagogical approaches at a high cognitive demand is facilitated by its selective student intake, the specific programmatic framework, and its dedicated resources. The Hector Seminar thus differs from typical regular school instruction practice in STEM, which is largely curriculum-driven, focusing on systematic concept development and foundational skills via rather teacher-guided approaches in larger, heterogeneous classes within shorter timeframes (45–90 min.). Although regular STEM education aspires to incorporate more extensive and interdisciplinary inquiry and project-based learning, frequently there is the argument that practical constraints, such as class size, curriculum breadth, limited time, and resource allocation, often hinder its widespread and sustained implementation (e.g., Teig et al., 2019).

Consequently, the emphasis in regular classrooms on systematic content coverage and foundational skills for a *diverse* student body, shaped by these prevalent constraints, clearly distinguishes it from the intensive, specialized, and project-oriented approach feasible within the resource-rich environment of the Hector Seminar.

Selection of mathematics as comparator subject

To compare the Hector Seminar to regular school teaching, we had to select an appropriate referent subject from the standard school curriculum. A comprehensive science subject, which might intuitively seem a close match for the interdisciplinary STEM nature of the Hector Seminar, is not uniformly implemented as a single, continuous discipline in the German secondary school system. While an integrated science subject covering biology, natural phenomena, and technology is taught in Grades 5 and 6, nothing comparable is anchored in the curriculum for later grades. Distinct science disciplines (e.g., physics, chemistry) are typically introduced from Grade 7 on. Elective science courses can be chosen from Grade 8 onwards, limiting their utility as a consistent comparator for all participants.

In contrast to this fragmented landscape for science subjects across the specific grades of our cohort, mathematics is the only core academic STEM subject that is taught universally and continuously to all students from Grade 5 onwards. Moreover, mathematics instruction is generally considered to be highly structured, and its curriculum is relatively standardized across different schools, thereby offering a more consistent educational experience for all students. Furthermore, its foundational role in STEM and its documented importance for the cognitive development and talent trajectories of gifted students (e.g., Preckel et al., 2020) underscore its pertinence. Finally, the Hector Seminar program emphasizes mathematical thinking and problem-solving skills, making it a domain explicitly valued within the enrichment program. Therefore, mathematics was chosen as the most appropriate and methodologically sound referent subject for this study.

Measures

Perceived teaching quality

Teaching quality was assessed for the three basic dimensions, with additional scales on adaptivity and interestingness of lectures, along with a rating of the perceived motivational climate of the two educational environments.

All students were asked to report on the perceived teaching quality provided by their regular math teacher ("My math teacher …") and provide a general evaluation of the motivational climate of their school experience (e.g., "My school…"). Hectorians were additionally asked to report on the teaching quality provided by their Hector Seminar teachers (e.g., "My Hector Seminar teachers…") and provide a general evaluation of the motivational climate of the Hector Seminar (e.g., "The Hector Seminar…"). Teaching quality scales originated from large German studies in the field of education, which were documented at the Research Data Center for Education at DIPF (FDZ Bildung, 2023). The motivational climate measure was self-developed on the basis of work by Heller (2009). Participants responded to all items on a 4-point rating scale ranging from 1 (*completely disagree*) to 4 (*completely agree*).

Three basic dimensions. *Effective classroom management* was assessed with five items (e.g., "…often has to intervene to ensure attention"; Baumert et al., 1997). *Cognitive activation* was assessed with seven items (e.g., "… often asks questions that one cannot answer spontaneously but that force you to think"; Baumert et al., 1997). *Student support* was assessed with seven items (e.g., "… offers me the opportunity to show what I can do"; Klieme et al., 2017). Complementing student support, *adaptivity* was assessed with seven items (e.g., "…responds to each

student individually by giving us tasks that match our ability"; Institut Für Qualitätsentwicklung Hessen, 2013). *Interestingness* was assessed with four items (e.g., "…presents us with interesting tasks"; Ditton & Merz, 2013).

Motivational climate. We assessed *motivational climate* with two self-developed items based on Heller (2009) for assessing students' general levels of interest (e.g., "Overall, the Hector Seminar is interesting and stimulating") and enjoyment (e.g., "School is fun for me"). Our items tap into the intrinsic value and general positive affect associated with the learning context. The conceptualization of motivational climate through student-perceived interest and enjoyment is grounded in Expectancy-Value Theory (EVT; Eccles & Wigfield, 2002), particularly its 'interest value' and 'attainment value' components, and prominent interest theories, such as the four-phase model of interest development (Hidi & Renninger, 2006). These theoretical frameworks suggest that learning environments that foster high levels of interest and enjoyment contribute to students' intrinsic motivation and engagement. The mean of the two items was used as a composite for the motivational climate in school, $r(324) = .66$, $p < .001$, and the Hector Seminar, $r(62) = .78$, $p < .001$.

Internal consistency was good for all scales, ranging from McDonald's $\omega = .60–.88$ (see Table 2 for all reliability estimates).

Covariates

Cognitive potential (in STEM). Participants were selected for the Hector Seminar on the basis of a ranking list derived within each cohort from a composite score of students' cognitive potential (in STEM). The score included students' general cognitive ability (i.e., intelligence) and STEM-related cognitive abilities (i.e., visual–spatial abilities, physical–technological problem-solving, and math abilities). This composite was based on all students undergoing the selection procedure in the respective year and was thus built on a larger database than the data analyzed in our study. Before the composite was built, scores from all scales used for the selection were *z*-standardized within each cohort to ensure comparability.

Intelligence. Students' fluid intelligence was assessed with the latest German adaptation of Thorndike's Cognitive Abilities Test (Kognitiver Fähigkeitstest III; KFT–III; Heller et al., in press; Heller & Perleth, 2000), which comprises nine subtests for assessing verbal, quantitative, and nonverbal reasoning abilities, with three scales each. However, the third scale of the nonverbal reasoning abilities (N3) was not assessed, as it was too similar to the unfolding test for assessing visual–spatial abilities. A composite score of the three subtests was used to capture students' intelligence. The reliability of the composite was satisfactory (McDonald's $\omega = .69$). Additionally, IQ was approximated by using the norm data from the short version of the KFT–III (Heller et al., in press). Based on this approximation, the mean of IQ was $M = 117.18$ ($SD = 9.3$) for non-Hectorians and $M = 133.7$ ($SD = 7.2$) for Hectorians.

STEM-related cognitive abilities. Students' visual–spatial abilities and physical-technological problem-solving were assessed with three scales from the Munich High Ability Test Battery (MHBT; Heller & Perleth, 2007). Visual–spatial abilities were assessed with a 25-item unfolding test and a 20-item mirror test. Physical-technological problem-solving was assessed with 15 items. A composite score comprising the three MHBT scales was used to capture students' STEM-related cognitive abilities. The reliability of the composite was found at the lower end of acceptability ($\omega = .56$).

Math ability. Students' math ability was assessed with a 13-item math ability test that was self-developed by the Hector Seminar teachers (Taulien et al., 2001). The items assessed mathematical knowledge and mathematical problem-solving skills, and the test has been shown to differentiate between gifted and nongifted students (Heller, 2009).

Selection composite. To calculate the overall placement composite score, the (composite) scores for intelligence, STEM-related cognitive abilities, and math ability were each *z*-standardized within cohorts and then assigned specific weights based on their importance (see Heller, 2009). Intelligence was weighted three times, math ability was weighted twice, and STEM-related cognitive abilities were weighted once. These weighted scores were then combined to create the final composite score ($\omega = .74$).

Demographics. Information on age, gender, and socioeconomic status (SES) was collected from students and parents. Parents' occupational status was used to compute the Highest International Socio-Economic Index of Occupational Status (HISEI; Ganzeboom et al., 1992) as the measure of SES. Descriptive statistics for these demographics and mean differences between Hectorians and non-Hectorians can be found in Table S1.

Significant gender differences were observed in Hector Seminar enrollment, with a higher proportion of boys participating, $\chi^2(1, N = 335) = 4.01$, $p = .045$, Cox index = -0.24 (see Table S1 for full details). As expected, and per definition, the two groups also differed on the selection criteria, with Hectorians showing greater values, $t(158.95) = 23.08$, $p < .001$, $d = 2.42$ (Table S1).

Table 2 Bivariate correlations between study variables and internal consistencies for scales

Variable	McDonald's ω	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1. Class. management (REG)	.60[a]	–																	
2. Cog. activation (REG)	.77	.32**	–																
3. Stud. support (REG)	.83	.44**	.45**	–															
4. Adaptivity (REG)	.77	.25**	.45**	.48**	–														
5. Interestingness (REG)	.83	.51**	.54**	.64**	.52**	–													
6. Motiv. climate (REG)	–	.23**	.28**	.25**	.22**	.32**	–												
7. Class. management (HS)	.63[a]	.17	.20	.14	.18	.24	.24	–											
8. Cog. activation (HS)	.78	−.19	.28*	.10	.28*	.19	.22	.21	–										
9. Stud. support (HS)	.88	−.15	−.04	.16	.03	−.02	.16	.41**	.47**	–									
10. Adaptivity (HS)	.60	−.09	.08	−.17	.07	.04	.12	.33*	.50**	.35**	–								
11. Interestingness (HS)	.77	−.11	.20	.05	.17	.25	.09	.41**	.37**	.49**	.43**	–							
12. Motiv. climate (HS)	–	.23	.33**	.31*	.23	.39**	.43**	.25*	.20	.18	.19	.32*	–						
13. Selection composite	.74	−.08	−.11	.06	.02	−.11*	−.14*	.05	.05	−.03	−.14	.10	.12	–					
14. Cognitive ability	.69	−.05	−.08	.08	.00	−.06	−.07	.08	.14	−.05	−.05	.07	.22	.85**	–				
15. Enrichment	–	−.04	−.04	.02	.00	−.08	−.11*	–	–	–	–	–	–	.70**	.53**	–			
16. Age	–	−.04	−.10	−.12*	−.11*	−.14*	−.15**	−.14	−.28*	−.22	−.16	−.15	−.39**	.05	.01	.01	–		
17. Gender	–	−.12*	−.03	−.21**	−.12*	−.10	.19**	.08	−.01	−.06	.11	−.11	−.02	−.10	−.06	−.07	.05	–	
18. SES	–	−.05	.03	−.02	.01	−.01	−.01	.12	.23	−.04	.12	.10	−.06	.17**	.13*	.08	−.16**	.09	–

REG regular school context, *HS* Hector Seminar, *Selection composite* STEM-related cognitive ability, *Gender* dichotomous (0 = *male*; 1 = *female*), *Enrichment* dichotomous (0 = *not attending Hector Seminar*; 1 = *attending Hector Seminar*), *SES* Socio-Economic Index of Occupational Status

[a] Item 1 from the effective classroom management scale was excluded due to low reliability

* $p < .05$. ** $p < .01$

Statistical analyses

Statistical analyses were performed using R (Version 4.4.1) with the *tidyverse* (Wickham et al., 2019), *lavaan* (Rosseel et al., 2024), and *rdd* (Calonico et al., 2014) packages. First, we examined demographic differences between Hectorians and non-Hectorians using independent-samples *t*-tests or the Chi-square test for binary variables (see Supplement Table S1). In Step 2, we analyzed mean differences in perceived teaching quality between the regular school context and the Hector Seminar for Hectorians. For this within-subject comparison, we used paired *t*-tests to account for the same participants providing ratings across both settings. In Step 3, we analyzed mean differences in perceived teaching quality in the regular school context between Hectorians and non-Hectorians, utilizing independent *t*-tests to capture between-group variations. To control for potential confounding of the selection composite, additional regression discontinuity analyses (Gelman & Hill, 2006) were calculated.

One-tailed tests were applied for our confirmatory, directed research question. In such cases, it is methodologically appropriate to test only for effects in the predicted direction, as it increases statistical power for the hypothesized direction, and maintains appropriate Type I error control given the pre-specified nature of the hypothesis (Hales, 2023). For our exploratory research question, we applied two-tailed tests. To determine effect sizes for these mean differences, Cohen's d was calculated. Magnitudes were interpreted based on empirical benchmarks proposed by Kraft (2020) for causal research on education interventions, with effect sizes interpreted as small ($d < 0.05$), medium ($d = 0.05$ to < 0.20), and large ($d \geq 0.20$) (Kraft, 2020). Finally, we used the Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995) to control the false discovery rate due to multiple testing within each research question.

Results

Descriptives and bivariate correlations

Tables 3 and 4 present descriptive statistics for the observed variables. Table 2 presents bivariate correlations between study variables.

As shown in Table 2, there were significant positive correlations between the six indicators of teaching quality within each respective educational context (except for cognitive activation's correlations with effective classroom management and motivational climate, as well as motivational climate with student support and adaptivity in the Hector Seminar). These findings can also be

Table 3 Comparison of teaching quality within Hectorians: regular school context and hector seminar

Variable	Hector seminar *M* (*SD*)	Regular math lesson *M* (*SD*)	*df*	t^a	Cohen's *d*
Class. management	3.31 (0.48)	2.68 (0.50)	51	7.27***	1.01
Cognitive activation	3.09 (0.59)	2.77 (0.59)	60	3.50***	0.45
Student support	3.05 (0.40)	3.17 (0.61)	61	−1.44 (*n.s.*)	−0.18
Adaptivity	2.25 (0.58)	2.26 (0.78)	58	−0.15 (*n.s.*)	−0.02
Interestingness	3.33 (0.59)	2.67 (0.76)	60	6.25***	0.80
Motivational climate	3.54 (0.60)	2.94 (0.71)	61	6.65**	0.85

Motivational climate refers to the general climate in the respective learning environment

[a] *p* values were corrected for multiple testing with the Benjamini–Hochberg procedure

*** $p < .001$ (one-tailed)

Table 4 Comparison of teaching quality in the regular school context between Hectorians and non-Hectorians

Variable	Hectorians *M* (*SD*)	Non-Hectorians *M* (*SD*)	*df*	t^a	Cohen's *d*
Class. management	2.61 (0.58)	2.67 (0.54)	321	−0.75 (*n.s.*)	−0.11
Cognitive activation	2.72 (0.65)	2.77 (0.53)	317	−0.63 (*n.s.*)	−0.09
Student support	3.13 (0.63)	3.11 (0.56)	324	0.31 (*n.s.*)	0.04
Adaptivity	2.22 (0.80)	2.22 (0.72)	324	0.00 (*n.s.*)	0.00
Interestingness	2.62 (0.80)	2.77 (0.74)	327	−1.39 (*n.s.*)	−0.19
Motivational climate	2.93 (0.71)	3.12 (0.67)	324	−2.03 (*n.s.*)	−0.28

Teaching quality was assessed in the context of students' regular math lessons. Motivational climate refers to the general climate in the school context

[a] *p* values (two-tailed) were corrected for multiple testing with the Benjamini–Hochberg procedure

interpreted as indicating the construct validity of these scales, as they were distinct but positively associated. Moreover, cognitive activation and motivational climate were also positively correlated across the educational contexts, pointing to a general tendency in how students perceived these teaching quality indicators. Interestingly, weak negative associations were found between students' STEM-related cognitive ability and the interestingness of their regular math lessons and the motivational climate of regular school, indicating systematic differences in students' perceptions of teaching quality, with students with higher STEM-related cognitive ability perceiving their regular math lessons as less interesting and the motivational climate of the regular school as less favorable. In addition, age and gender effects were observed, with older students and those identifying as female tending to perceive lower teaching quality.

Comparison of perceived teaching quality: in- and out-of-school contexts

To compare the perceived teaching quality of the two educational contexts, within-subject comparisons using paired *t*-tests were computed for each outcome variable (Table 3).

The results revealed significant differences in teaching quality between the Hector Seminar and regular schooling for perceived effective classroom management (*H1a*), cognitive activation (*H1b*), interestingness (*H1e*), and motivational climate (*H1f*), with large effect sizes according to the benchmarks for educational interventions proposed by Kraft (2020). No differences were found for perceived instructional support (*H1c*) or adaptivity (*H1d*). Thus, Hypothesis 1 was partly supported.

Group differences in perceived teaching quality in regular schooling between Hectorians and non-Hectorians

Independent *t*-tests were computed for each teaching quality scale to compare the perceived teaching quality in regular schooling between Hectorians and non-Hectorians. Table 4 presents the results.

There were no significant differences in perceived teaching quality in regular schools between Hectorians and non-Hectorians (Table 4). To ensure that these findings were not confounded by the selection criteria used for the group assignment, additional regression discontinuity analyses were conducted, thereby strengthening the validity of our findings.

Controlling for selection criteria: regression discontinuity analyses

Ideally, regression discontinuity designs (RDD) require a sharp cutoff where assignment to the treatment group (e.g., Hector Seminar) is determined solely by whether an individual's score on a predetermined assignment variable falls above or below a specific threshold. However, our study features a fuzzy RDD, meaning that assignment to the Hector Seminar was not perfectly determined by a cutoff in the selection composite score. This fuzziness arises from several factors. Firstly, the Hector Seminar operates across four locations, where the best 10% of students are meant to be selected via the composite score in each location, thus slight deviations in the cutoff of the composite score between locations are possible. Secondly, even when offered a place in the program, some students might decline participation for various personal reasons. Consequently, a cutoff had to be determined in the current study.

The cutoff for the regression discontinuity analyses was set at a composite value of 60. We chose this threshold for several reasons: first, it represented a point where the assignment to the treatment group increased notably, ensuring practical relevance. Second, it ensured a sufficient number of observations on both sides of the threshold, which was essential for statistical balance and reliable estimation. Visual inspection of the data further supported the choice of this cutoff, as it provided a clear separation of students attending the Hector Seminar. Only one Hectorian was below this cutoff with a value of 58. Twenty-three Non-Hectorians were above the cutoff.

This deviation from a strict assignment rule necessitates a careful examination of the distribution of the assignment variable around the cutoff. Consequently, we used the cutoff to perform a density discontinuity test to verify the assumption of a continuous distribution of the running variable at the cutoff (McCrary, 2008). Ensuring density continuity is crucial for validating that groups on either side of the cutoff are comparable and that observed differences are not due to selection effects but rather to the intervention assignment. The corresponding density plot (Fig. 2) visualizes the distribution of the selection composite scores relative to the cutoff. The results

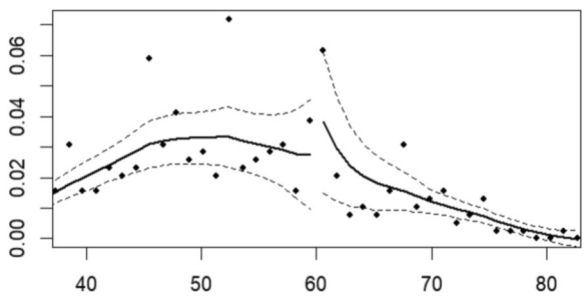


Fig. 2 RDD density plot. The solid black line represents the estimated density; the dashed lines depict the confidence intervals. The cutoff was set to 60. There was no significant change in the density of the selection composite scores at the cutoff ($p = .127$)

indicated no significant change in the density of the selection composite scores at the cutoff ($p = .127$), supporting the assumption of continuity in the running variable at the threshold and thereby affirming the validity of the RDD approach in this context.

The apparent visual discontinuity illustrated in Fig. 2 can be attributed to the smoothing parameters used in the density estimation. Specifically, the data-driven and automatically selected bandwidth (9.68) and bin size (1.16) balance smoothness and detail but may exaggerate minor fluctuations in the density curve. Furthermore, the confidence intervals encompass the observed variation at the cutoff, supporting the conclusion of continuity in the underlying data distribution.

In the next step, we scanned the scatterplots to visually examine potential discontinuities before testing them.

Overall, no linear patterns or visual discontinuities were detected (Fig. 3). Finally, regression discontinuity analyses supported the findings from the group comparison tests (Table 5) and scatterplots.

Overall, the results shown in Table 5 revealed no significant discontinuity at the selection cutoff. Conclusively, there were no significant differences in the perceived teaching quality of the regular school context between students who participated in the Hector Seminar and students who did not, even when we controlled for the selection composite.

Robustness checks

To check for the robustness of our results across different comparison approaches, we ran two additional analyses. First, we analyzed a subsample of students with a sharp RDD. To this end, we excluded all nonmatching students from above the cutoff and the one from below (e.g., all students with a selection composite above the cutoff who were not attending the Hector Seminar were excluded from the regression discontinuity analyses) and reran the analyses. The analyses yielded the same results (see Supplement Figure S1 and Table S2). As an additional robustness check, we included only students who were above the selection cutoff, Hectorians and non-Hectorians, and compared their perceptions of the teaching quality in their regular classes (Supplement Table S2). Taken together, these additional analyses confirmed the robustness of our results.

Discussion

The purpose of the current study was to evaluate the perceived teaching quality of a specialized out-of-school STEM enrichment program designed for highly gifted students and compare it with the perceived teaching quality of their regular school setting. The study thus built upon previous teaching quality research, which has traditionally focused on regular classroom settings (e.g., Wagner et al., 2016). Our findings revealed that gifted

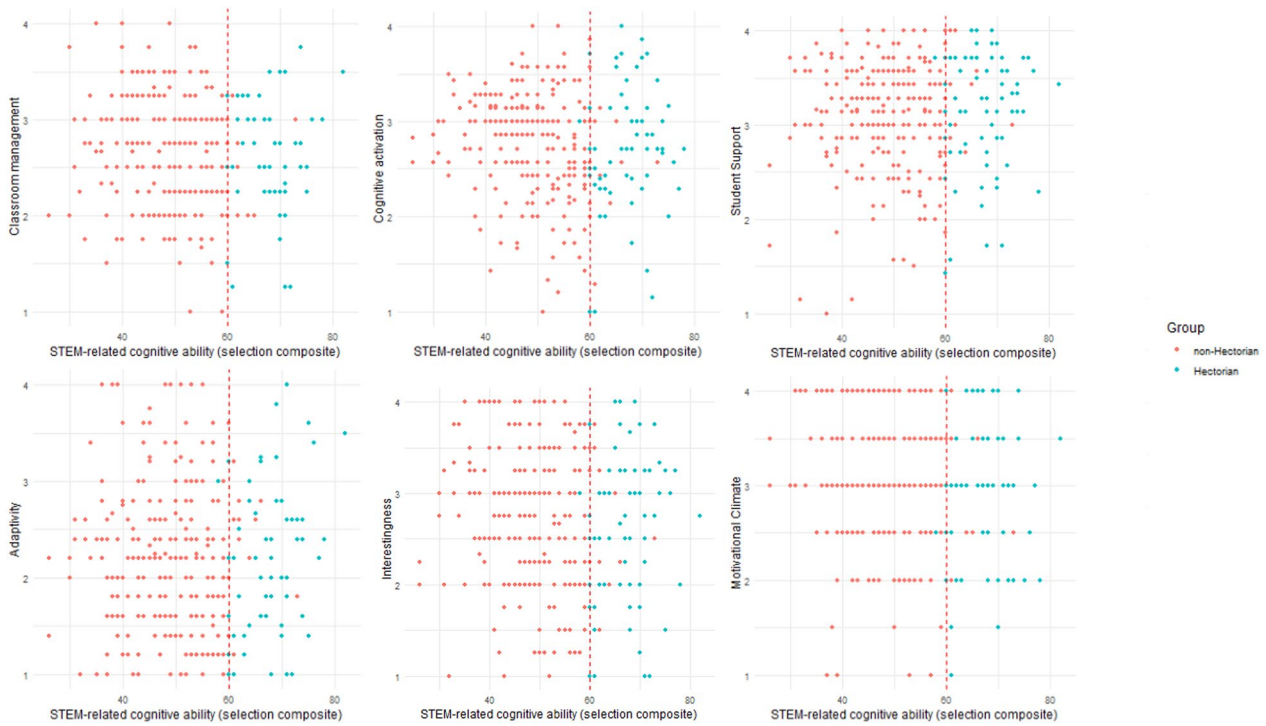


Fig. 3 Fuzzy regression discontinuity designs for teaching quality

Table 5 Regression discontinuity analyses for (teaching) quality scales at the cutoff of 60

Effect	Effective classroom management				Cognitive activation				Student support			
	B	*SE*	95% CI	p^a	*B*	*SE*	95% CI	p^a	*B*	*SE*	95% CI	p^a
(Intercept)	2.60	0.06	[2.48, 2.71]	< .001	2.65	0.06	[2.54, 2.77]	< .001	3.15	0.06	[3.03, 3.27]	< .001
Enrichment	−0.07	0.16	[−0.38, 0.25]	.761	−0.04	0.14	[−0.31, 0.23]	.761	−0.39	0.14	[−0.39, 0.16]	.633
Selection composite	−0.01	0.00	[−0.01, 0.00]	.122	−0.01	0.00	[−0.02, 0.00]	.014	0.00	0.00	[0.00, 0.01]	.393
Enrichment x selection	0.02	0.02	[−0.02, 0.05]	.332	0.03	0.01	[0.00, 0.05]	.086	0.02	0.01	[−0.02, 0.04]	.540
R^2	.01				.02				.01			
Effect	**Adaptivity**				**Interestingness**				**Motivational climate**			
	B	*SE*	95% CI	p^a	*B*	*SE*	95% CI	p^a	*B*	*SE*	95% CI	p^a
(Intercept)	2.21	0.08	[2.06, 2.35]	< .001	2.65	0.08	[2.50, 2.80]	< .001	3.03	0.07	[2.89, 3.17]	< .001
Enrichment	−0.33	0.18	[−0.68, 0.01]	.353	−0.19	0.18	[−0.55, 0.17]	.633	−0.15	0.16	[−0.48, 0.17]	.633
Selection composite	0.00	0.01	[−0.01, 0.01]	.773	−0.01	0.01	[−0.02, 0.00]	.058	−0.01	0.01	[−0.02, 0.00]	.100
Enrichment x Selection	0.05	0.02	[0.01, 0.08]	.009	0.03	0.02	[0.00, 0.07]	.087	0.02	0.02	[−0.02, 0.05]	.350
R^2	.02				.02				.02			

The selection composite was mean-centered at the cutoff point of 60

[a] The enrichment predictor variable's *p* values were corrected for multiple testing with the Benjamini–Hochberg procedure

students generally evaluated the teaching quality of the extracurricular STEM enrichment program we investigated (i.e., the Hector Seminar) as significantly higher compared with their regular school context.

However, one exception was the dimension of student support, for which there were no significant differences in perceived teaching quality, particularly for the aspect of the adaptivity of the teaching. Moreover, there was no significant difference in the perceived teaching quality of the regular classes between the students who attended the Hector Seminar and those who did not. The findings indicate the potential of specialized programs to better meet the unique learning needs of gifted students by offering tailored and stimulating instruction.

Perceived teaching quality of specialized out-of-school STEM enrichment

Students attending the Hector Seminar evaluated the cognitive activation, effective classroom management, interestingness, and motivational climate of the Hector Seminar as higher than in their regular school classes. Specifically, the large magnitude of these effect sizes indicates not only statistical significance, but also points to their practical relevance in terms of educationally meaningful advantages in teaching quality for students attending the Hector Seminar. Overall, these results emphasize that specialized out-of-school enrichment programs that are designed to enhance and deepen learning by addressing the unique aptitudes, interests, and needs of gifted learners (Reis & Renzulli, 2023) indeed provide a (perceived) cognitively activating, interesting, and enhancing learning environment for gifted learners.

This study focused on well-known teaching quality dimensions, which are recognized for providing an efficient description of effective classroom learning. The ways in which teachers achieve these different quality dimensions are manifold. Thus, there might be several reasons for the higher perceived teaching quality in the enrichment program. However, as this study focused on cross-sectional differences, we can only speculate about the underlying causes as follows.

First, as theorized, in enrichment programs, teachers might be able to tailor their teaching better, for example, with highly demanding learning materials and learning content that goes beyond the curriculum, presented at a faster pace and differentiated at the individual level (Tieso, 2003). However, in this study, nonsignificant differences between the teaching quality of the Hector Seminar and the regular school context were found in the comparatively more teacher-centered dimensions of student support, especially adaptivity.

On one hand, it is plausible that the Hector Seminar's macrolevel adaptation and inquiry-based pedagogy inherently offer a form of structural adaptivity that students might not recognize as 'teacher adaptivity' in the traditional sense. For instance, the emphasis on open-ended, inquiry-based tasks and project work naturally allows students to engage with material at their own pace and depth. On the other hand, anecdotal reports from Hector Seminar instructors involve intentionally providing less direct individual feedback, instead encouraging gifted students to engage in deeper self-reflection and learn from their own problem-solving efforts and mistakes.

This result indicates that the reason for the higher perceived teaching quality might be more likely to be found in the program's macro adaptation (e.g., the setup and format of the enrichment program) than in teachers' abilities to micro adapt (Hardy et al., 2019). These macro-adaptations include the fundamental setup of the program, such as the selection of highly able and motivated students and the flexible program structure. It appears that these overarching structural and contextual features of the enrichment environment itself are what students perceive as particularly different, perhaps more so than specific in-the-moment instructional adjustments by individual teachers.

A frequent distinction in this context is between direct instruction and learning formats that require students to actively contribute to the lesson (e.g., inquiry-based learning). One might assume that students in out-of-school programs are particularly likely to benefit from active learning formats, as these programs often allow for greater flexibility and a stronger focus on student-centered approaches (Reis & Renzulli, 2023).

Furthermore, the Hector Seminar's specific resources and programmatic framework targeting gifted students (Heller, 2009) facilitate the implementation of cognitively activating and interesting tasks precisely matched to their abilities, which might lead to higher ratings of teaching quality. The format of the program could thereby provide an additional explanation of why teaching quality in out-of-school programs is perceived more favorably, as these formats may be better aligned with gifted students' learning preferences and cognitive needs (Reis & Renzulli, 2023). This idea is in line with the notions that teaching quality is not limited to teacher behavior, but also depends on student behavior (Göllner et al., 2020) and that ability grouping effects might also be due to institutional and structural benefits (Pallas et al., 1994).

Additionally, students in out-of-school programs often experience academically oriented social networks that facilitate the spread of academic norms and higher academic aspirations (Göllner et al., 2020). Moreover, teachers might spend less time dealing with disciplinary problems and might devote more time to instruction (Tieso, 2003). For example, gifted students are often well-adjusted and highly interested (Reis & Renzulli, 2004), so there are likely fewer disruptions by students in the Hector Seminar, resulting in less need for effective classroom management, which in turn might lead to perceptions of more effective classroom management.

When interpreting the higher perceived teaching quality in the Hector Seminar, it is also important to consider the potential role of teacher selection and characteristics. Specialized enrichment programs for gifted students may be in a position to attract or recruit the most effective and experienced teachers with strong motivation for working with this student population in a less restrictive curricular environment. However, disentangling the effects of teacher expertise from program structure remains a task for future research with more direct comparisons of teacher profiles.

Synthesizing these points, the emphasis on active learning formats and student autonomy in enrichment programs like the Hector Seminar might inherently reshape the nature of teacher support. While fostering independence and inquiry, such approaches may offer fewer instances of direct teacher guidance or individualized feedback compared to more structured regular schooling. This shift in pedagogical dynamics, with teachers acting more as facilitators, could explain why students perceived a similar level of overall student support across both contexts, despite other notable differences in teaching quality.

Regardless of the underlying reasons, the study clearly demonstrates that students perceive that the teaching quality in out-of-school programs is higher than that in regular school contexts. This finding is particularly significant, given the established link between perceived teaching quality and positive learning outcomes (e.g., Wagner et al., 2016). Our findings underscore the value of additional learning opportunities for gifted students and support initiatives aimed at providing specialized STEM enrichment programs.

Reference effects: differences in perceived teaching quality in regular schooling

Our findings showed that although the students who attended the Hector Seminar perceived higher teaching quality at the seminar, they did not perceive the teaching quality in their regular classes differently compared to their peers. First, this outcome is positive in indicating that at least students who experienced this specialized enrichment program did not show a huge deviation from the baseline set by their peers who did not attend the Hector Seminar, even when we controlled for selection criteria.

Several interpretations that are not mutually exclusive may explain this absence of a discernible group difference. First, the perceptions of regular school teaching might be particularly strong and stable, especially among gifted students who may possess a refined ability to differentiate between distinct learning contexts and their respective purposes. Second, it is possible that opposing comparison or reference effects were simultaneously at play (e.g., BFLP and ingroup–outgroup comparison), effectively canceling each other out. Third, the magnitude of any specific reference effect attributable to the extracurricular program might simply have been too subtle to

produce statistically significant group differences within our sample when considering the broader, more consistently experienced regular school environment. Fourth, significant individual variability in students' processing of cross-contextual experiences (Göllner et al., 2018) could have led to varied individual responses that, in total, resulted in no net group-level difference.

However, as these results are cross-sectional and descriptive, no causal conclusion on how attending the Hector Seminar influences perceptions of regular teaching is possible. The question that remains is whether students' perceptions of regular teaching are influenced by attending the Hector Seminar. Bivariate correlations indicated that higher STEM-related cognitive potential was negatively associated with perceived cognitive activation and interestingness in regular teaching. This suggests that highly able students generally tend to rate these aspects lower than their less able peers do. Thus, it is unlikely that students' perceptions decreased after attending the Hector Seminar; yet, there was no significant deviation from their peers who did not attend the seminar.

Taken together, the study provides the first descriptive insights into potential reference and comparison effects. At least students attending the Hector Seminar did not perceive the teaching quality in their regular classes as lower than their peers did. However, future research should involve longitudinal investigations of potential change to be able to make precise statements.

Strengths and limitations

This study is the first to compare teaching quality between in- and out-of-school STEM learning environments. A major strength is the innovative research question, which is an important first step toward a better understanding of specialized learning environments. Furthermore, the research design allowed us to conduct a comprehensive comparison of students' perceptions of teaching quality between in- and out-of-school programs, applying robust research methods to account for the challenging design and to ensure comparability between the two contexts. However, several potential limitations should be kept in mind when interpreting our findings.

First, the present study was based on cross-sectional data, which provides initial descriptive insights into differences between in- and out-of-school STEM teaching. However, it does not allow conclusions about underlying causes or the causality of observed effects. Despite this limitation, our findings represent a valuable starting point for further investigation.

Second, our results relied on subjective evaluations. Although these evaluations have been shown to be reliable predictors of learning outcomes and are thus worthy of consideration (Lüdtke et al., 2007), they are not entirely free from perception biases and do not represent true, unbiased teaching quality values. Results might be biased due to perception biases such as labeling effects, the halo effect, or classroom composition features (e.g., Coleman et al., 2015; Göllner et al., 2020). For example, social desirability might bias students' answers because of societal norms or stereotypes suggesting gifted programs' superiority (Bou Malham & Saucier, 2016). The Hector Seminar constitutes one such highly prestigious program. However, the different patterns of findings in single subdimensions speak against perception biases, as they should affect all the teaching quality scales to similar extents. Additionally, student ratings can be influenced by how questions are framed (Göllner et al., 2020). For example, Jaekel et al. (2022) found that differences in associations depended on the referents (e.g., "me" or "us"); however, such differences were small. A promising solution for mitigating these issues in future research could be to use observational data to complement self-reported measures.

Third, we compared the teaching in the Hector Seminar, which is provided in STEM subjects by a tandem of teachers, with regular mathematics instruction provided by a single teacher. This involves non-equivalence in both subject matter scope and teaching situation, which are important considerations when interpreting our findings. The broader, interdisciplinary nature of the interdisciplinary STEM content within the enrichment program, potentially offering more novelty and topic variety, may inherently lead to different student perceptions compared to the more focused and sequential mathematics curriculum.

Furthermore, the prevalent team-teaching model in the Hector Seminar, contrasting with typical single-teacher mathematics instruction, could contribute to differences in perceived support, classroom dynamics, or instructional variety. Disentangling these effects from the core program elements requires further research. However, whereas this comparison did not provide perfect comparability, it represented the closest possible contrast and served as a solid first step toward gaining valuable insights. Additionally, the three core dimensions of teaching quality are considered genuine quality dimensions that are valid across academic subjects (e.g., STEM subjects) and should, therefore, be sufficiently comparable.

Implications and future research

Focusing on the evaluation of the teaching quality of specialized STEM enrichment programs, this study ultimately contributes to a better understanding of how (gifted) students perceive teaching quality in diverse STEM learning environments. The study highlights the potential of specialized STEM enrichment programs to

address the needs of gifted learners and contribute to their educational development. These insights offer several implications for practice and future research and should be used to further improve (gifted) education.

First, enrichment programs can serve as models for improving teaching quality, particularly in providing cognitively activating and engaging learning environments. For example, incorporating challenging, inquiry-based tasks and project-based learning (Blanchard et al., 2010; Sahin, 2013) might enhance the learning experience for (gifted) students in regular classrooms. Policymakers could support this by encouraging curriculum frameworks that allow for greater depth and autonomy for advanced learners.

Whereas replicating the exact selectivity of extracurricular programs in every classroom is not feasible, policymakers should consider how elements of this macro adaptation design can be implemented. This could involve supporting schools in creating specialized STEM tracks, advanced learning groups, or dedicated project times within the regular school day that allow for a more focused and stimulating environment for gifted learners.

Moreover, there are good reasons to assume that teachers emphasize autonomy, and inquiry-based learning is one key element. However, a more fine-grained investigation of the specific learning formats, including their temporal dynamics, is necessary to clarify how these features influence teaching quality in extracurricular settings. Future research should assess potential learning outcomes to investigate whether improvements are associated with higher levels of teaching quality in the extracurricular learning environment. In addition, the role of the instructional methods used should be investigated further, particularly regarding its effects on teaching quality ratings.

A critical direction for future research involves investigating whether established core dimensions of teaching quality, such as student support, hold the same meaning and entail the same optimal practices for gifted students as they do in general education contexts. It is conceivable that effective teaching for gifted students in an enrichment setting manifests differently than captured by standard conceptualizations or measures, as anecdotally reported by the Hector Seminar teachers, who sometimes intentionally provide less direct individual feedback.

Additionally, future research should incorporate longitudinal data to analyze potential changes in perceived teaching quality and whether reference and comparison effects are responsible for these potential changes. Furthermore, including older cohorts from the Hector Seminar would allow for an investigation of developmental trajectories and the impact of increased experience on students' perceptions. In addition, further studies should examine whether the findings generalize to nongifted educational settings.

Moreover, the results also revealed practical limitations, such as the overall insufficient focus on individualized teaching, showing that independent of the learning context, teachers have difficulties individualizing their teaching (Richards & Omdal, 2007). Student support and adaptivity require teachers to have good content knowledge to know their students' levels of prior knowledge, interest, and motivation (Corno, 2008). Along with the perceptions of the enrichment program as highly interesting, the results underscore the need for targeted professional development for teachers in STEM education (Huang et al., 2022; A. Kim, 2025).

Teachers are expected to develop their classroom practices in response to students with an increasingly diverse set of individual backgrounds and educational needs (OECD, 2016), including gifted students (KMK, 2016). Policymakers should consider scaling such programs to ensure equitable access for gifted students. Structured programs focusing on specialized enrichment, such as the Schoolwide Enrichment Model (Reis & Renzulli, 2023), might be the first steps toward a better way to implement gifted education.

Finally, the systematic variance in teaching quality evaluations found in negative correlations between students' STEM-related cognitive ability and interestingness and motivational climate suggests that gifted students might perceive and evaluate teaching quality slightly differently than their nongifted peers.

Conclusion

The current study provides the first valuable insights into the teaching quality of specialized learning environments and how they are perceived by gifted students in comparison with regular school contexts. The findings show that additional specialized STEM enrichment programs constitute high-quality learning environments for gifted students. The overall format of the program, as macrolevel adaptation, seems to be a key factor in this regard. Moreover, students attending the Hector Seminar do not perceive the teaching in their regular classes as lower than their peers who did not attend the seminar. These findings contribute to a more nuanced understanding of gifted education and STEM teaching quality in diverse learning environments and underscore the value of enrichment programs in gifted education.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40594-025-00576-w.

Supplementary material 1.

Acknowledgements
We thank Amelie Schönle for her support in data collection, Franziska Rahm for her support with scoring, and Jane Zagorski for language editing. We acknowledge support from the Open Access Publication Fund of the University of Tübingen.

Author contributions
A.-K.J.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, visualization, writing—original draft (lead) W.W.: methodology, writing—review and editing T.F.: writing – review and editing R.G.: writing—review and editing U.T.: conceptualization, funding acquisition, supervision, writing—original draft (supporting), writing—review and editing.

Funding
Open Access funding enabled and organized by Projekt DEAL. This research was supported by the Hector Foundation and by the Postdoctoral Academy of Research on Education (PACE), funded by the Baden-Württemberg Ministry of Science, Research, and the Arts, at the Hector Research Institute of Education Sciences and Psychology, Tübingen.

Data availability
All data, analysis codes, and research materials are available at https://osf.io/4y6qp/files.

Declarations

Ethics approval and consent to participate
The Ethics and Data Protection Commission of the University of Tübingen approved the study (A2.5.4-239.3_hb). All participants provided written informed consent after being fully informed about the study procedures and their right to withdraw.

Competing interests
The authors declare no competing interests.

Received: 31 December 2024 Accepted: 19 September 2025
Published online: 06 October 2025

References

Arens, A. K., & Möller, J. (2016). Dimensional comparisons in students' perceptions of the learning environment. *Learning and Instruction, 42*, 22–30. https://doi.org/10.1016/j.learninstruc.2015.11.001

Bardach, L., Lüftenegger, M., Yanagida, T., Spiel, C., & Schober, B. (2019). Achievement or agreement – Which comes first? Clarifying the temporal ordering of achievement and within-class consensus on classroom goal structures. *Learning and Instruction, 61*, 72–83. https://doi.org/10.1016/j.learninstruc.2019.01.003

Baumert, J., Gruehn, S., Heyn, S., Köller, O., & Schnabel, K. (1997). *Bildungsverläufe und psychosoziale Entwicklung im Jugendalter (BIJU) Dokumentation, Skalen Längsschnitt I, Welle 1–4* (Vol. 1). Max-Planck-Institut für Bildungsforschung.

Baumert, J., Kunter, M., Blum, W., Brunner, M., Voss, T., Jordan, A., Klusmann, U., Krauss, S., Neubrand, M., & Tsai, Y.-M. (2010). Teachers' mathematical knowledge, cognitive activation in the classroom, and student progress. *American Educational Research Journal, 47*(1), 133–180. https://doi.org/10.3102/0002831209345157

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B: Statistical Methodology, 57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Blanchard, M. R., Southerland, S. A., Osborne, J. W., Sampson, V. D., Annetta, L. A., & Granger, E. M. (2010). Is inquiry possible in light of accountability?: A quantitative comparison of the relative effectiveness of guided inquiry and verification laboratory instruction. *Science Education, 94*(4), 577–616. https://doi.org/10.1002/sce.20390

Bou Malham, P., & Saucier, G. (2016). The conceptual link between social desirability and cultural normativity. *International Journal of Psychology, 51*(6), 474–480. https://doi.org/10.1002/ijop.12261

Calonico, S., Cattaneo, M., Farrell, M., & Titiunik, R. (2014). *rdrobust: Robust Data-Driven Statistical Inference in Regression-Discontinuity Designs* (Version 2.2) [Computer software]. https://CRAN.R-project.org/package=rdrobust

Christ, A. A., Capon-Sieber, V., Köhler, C., Klieme, E., & Praetorius, A.-K. (2024). Revisiting the three basic dimensions model: A critical empirical investigation of the indirect effects of student-perceived teaching quality on student outcomes. *Frontline Learning Research, 12*(1), 66–123. https://doi.org/10.14786/flr.v12i1.1349

Coleman, L., Micko, K., & Cross, T. (2015). Twenty-five years of research on the lived experience of being gifted in school: Capturing the students' voices. *Journal for the Education of the Gifted, 38*(4), 1–19. https://doi.org/10.1177/0162353215607322

Corno, L. (2008). On teaching adaptively. *Educational Psychologist, 43*(3), 161–173. https://doi.org/10.1080/00461520802178466

Decristan, J., Fauth, B., Kunter, M., Büttner, G., & Klieme, E. (2017). The interplay between class heterogeneity and teaching quality in primary school. *International Journal of Educational Research, 86*, 109–121. https://doi.org/10.1016/j.ijer.2017.09.004

Ditton, H., & Merz, D. (2013). *QuaSSU - QualitätsSicherung in Schule und Unterricht—Erhebungszeitpunkt 2 (Skalenkollektion)*. https://doi.org/10.7477/18:31:1

Dumont, H. (2019). Neuer Schlauch für alten Wein? Eine konzeptuelle Betrachtung von individueller Förderung im Unterricht. *Zeitschrift Für Erziehungswissenschaft, 22*(2), 249–277. https://doi.org/10.1007/s11618-018-0840-0

Eccles, J. S., & Wigfield, A. (2002). Motivational beliefs, values, and goals. *Annual Review of Psychology, 53*(1), 109–132. https://doi.org/10.1146/annurev.psych.53.100901.135153

Fauth, B., Wagner, W., Bertram, C., Göllner, R., Roloff, J., Lüdtke, O., Polikoff, M. S., Klusmann, U., & Trautwein, U. (2020). Don't blame the teacher? The need to account for classroom characteristics in evaluations of teaching quality. *Journal of Educational Psychology, 112*(6), 1284–1302. https://doi.org/10.1037/edu0000416

Fütterer, T., Hoch, E., & Dumont, H. (2024). *Uncovering the relationship between technology-enhanced, adaptive teaching and situational interest in mathematics in a randomized controlled trial.* Research Square. https://doi.org/10.21203/rs.3.rs-5238796/v1

Fütterer, T., Hoch, E., Lachner, A., Scheiter, K., & Stürmer, K. (2023). High-quality digital distance teaching during COVID-19 school closures: Does familiarity with technology matter? *Computers & Education*. https://doi.org/10.1016/j.compedu.2023.104788

Ganzeboom, H. B. G., De Graaf, P. M., & Treiman, D. J. (1992). A standard international socio-economic index of occupational status. *Social Science Research, 21*(1), 1–56. https://doi.org/10.1016/0049-089X(92)90017-B

Gelman, A., & Hill, J. (2006). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press. https://doi.org/10.1017/CBO9780511790942

Göllner, R., Fauth, B., Lenske, G., Praetorius, A.-K., & Wagner, W. (2020). Do student ratings of classroom management tell us more about teachers or about classroom composition? *Zeitschrift Für Pädagogik Beiheft, 1*, 156–172. https://doi.org/10.3262/ZPB2001156

Göllner, R., Wagner, W., Eccles, J. S., & Trautwein, U. (2018). Students' idiosyncratic perceptions of teaching quality in mathematics: A result of rater tendency alone or an expression of dyadic effects between students and teachers? *Journal of Educational Psychology, 110*(5), 709–725. https://doi.org/10.1037/edu0000236

Gomez-Arizaga, M. P., Valdivia-Lefort, M., Castillo-Hermosilla, H., Hébert, T. P., & Conejeros-Solar, M. L. (2020). Tales from within: Gifted students' lived experiences with teaching practices in regular classrooms. *Education Sciences, 10*(5), 137. https://doi.org/10.3390/educsci10050137

Hachfeld, A., & Lazarides, R. (2021). The relation between teacher self-reported individualization and student-perceived teaching quality in linguistically

heterogeneous classes: An exploratory study. *European Journal of Psychology of Education, 36*(4), 1159–1179. https://doi.org/10.1007/s10212-020-00501-5

Hales, A. H. (2023). One-tailed tests: Let's do this (responsibly). *Psychological Methods*. https://doi.org/10.1037/met0000610

Hamre, B., Pianta, R., Downer, J., DeCoster, J., Mashburn, A., Jones, S., Brown, J., Cappella, E., Atkins, M., Rivers, S., & Hamagami, A. (2013). Teaching through interactions: Testing a developmental framework of teacher effectiveness in over 4,000 classrooms. *The Elementary School Journal, 113*, 461–487. https://doi.org/10.1086/669616

Hardy, I., Decristan, J., & Klieme, E. (2019). Adaptive teaching in research on learning and instruction. *Journal for Educational Research Online*. https://doi.org/10.25656/01:18004

Harker, R., & Tymms, P. (2004). The effects of student composition on school outcomes. *School Effectiveness and School Improvement*. https://doi.org/10.1076/sesi.15.2.177.30432

Heller, K. A. (2009). *Das Hector-Seminar: Ein wissenschaftlich evaluiertes Modell der Begabtenförderung im MINT-Bereich*. LIT Verlag. https://www.lit-verlag.de/isbn/978-3-643-10234-8

Heller, K. A., & Perleth, C. (2000). *Kognitiver Fähigkeitstest für 4. Bis 12. Klassen, Revision: KFT 4–12+ R*. Beltz Test. https://www.testzentrale.de/shop/kognitiver-faehigkeitstest-fuer-4-bis-12-klassen-revision.html

Heller, K. A., & Perleth, C. (2007). *MHBT-S - Münchner Hochbegabungstestbatterie für die Sekundarstufe*. Hogrefe. https://www.testzentrale.de/shop/muenchner-hochbegabungstestbatterie-fuer-die-sekundarstufe.html

Heller, K. A., Perleth, C., Gherairi, J., & Stumpf, E. (in press). *Kognitiver Fähigkeitstest III (KFT-III). Ein Verfahren für die Klassenstufen 4–12+*. Hogrefe.

Helmke, A. (2007). *Unterrichtsqualität und Unterrichtsentwicklung: Wissenschaftliche Erkenntnisse zur Unterrichtsforschung und Konsequenzen für die Unterrichtsentwicklung*. Bertelsmann.

Institut Für Qualitätsentwicklung Hessen. (2013). *Fragebogenerhebung Hessischer Referenzrahmen Schulqualität (Skalenkollektion)*. https://doi.org/10.7477/44:48:1

Hidi, S., & Renninger, K. A. (2006). The four-phase model of interest development. *Educational Psychologist, 41*(2), 111–127. https://doi.org/10.1207/s15326985ep4102_4

Holzberger, D., & Prestele, E. (2021). Teacher self-efficacy and self-reported cognitive activation and classroom management: A multilevel perspective on the role of school characteristics. *Learning and Instruction*. https://doi.org/10.1016/j.learninstruc.2021.101513

Huang, B., Siu-Yung Jong, M., Tu, Y.-F., Hwang, G.-J., Chai, C. S., & Yi-Chao Jiang, M. (2022). Trends and exemplary practices of STEM teacher professional development programs in K-12 contexts: A systematic review of empirical studies. *Computers & Education, 189*, 104577. https://doi.org/10.1016/j.compedu.2022.104577

Jaekel, A.-K., Göllner, R., & Trautwein, U. (2020). How students' perceptions of teaching quality in one subject are impacted by the grades they receive in another subject: Dimensional comparisons in student evaluations of teaching quality. *Journal of Educational Psychology, 113*(4), 770–783. https://doi.org/10.1037/edu0000488

Jaekel, A.-K., Wagner, W., Trautwein, U., & Göllner, R. (2022). "The teacher motivates us—or me?"—The role of the addressee in student ratings of teacher support. *Contemporary Educational Psychology*. https://doi.org/10.1016/j.cedpsych.2022.102120

Jaggy, A.-K., John, D., Wytopil, N., Schönle, A., Golle, J., Nagengast, B., & Trautwein, U. (2025). Talent-Study—Pre-registration of the Study Design [Preregistration *OSF*. https://doi.org/10.17605/OSF.IO/KBWZ7

Kahveci, N. Gü., & Akgül, S. (2014). Gifted and talented students' perceptions on their schooling: A survey study. *Gifted and Talented International, 29*(1–2), 79–91. https://doi.org/10.1080/15332276.2014.11678431

Kazak, A. E. (2018). Editorial: Journal article reporting standards. *American Psychologist, 73*(1), 1–2. https://doi.org/10.1037/amp0000263

Khalil, M., & Accariya, Z. (2016). Identifying "good" teachers for gifted students. *Creative Education, 7*(3), 407–418. https://doi.org/10.4236/ce.2016.73040

Kim, A. (2025). Impact of systematic support in teacher education and professional development on training-teaching alignment and instructional quality. *Journal for STEM Education Research*. https://doi.org/10.1007/s41979-025-00154-3

Kim, M. (2016). A meta-analysis of the effects of enrichment programs on gifted students. *Gifted Child Quarterly, 60*(2), 102–116. https://doi.org/10.1177/0016986216630607

Klieme, E., Fischer, N., Decristan, J., Sauerwein, M., Theis, D., Wolgast, A., & Züchner, I. (2017). Fragebogenerhebung StEG-S - 2. Erhebungswelle (Skalenkollektion). *Studie Zur Entwicklung von Ganztagsschulen - Teilstudie S*. https://doi.org/10.7477/199:217:1

Klieme, E., Pauli, C., & Reusser, K. (2009). The Pythagoras Study. Investigating effects of teaching and learning in Swiss and German mathematics classrooms. In T. Janik & T. Seidel (Eds.), *The power of video studies in investigating teaching and learning in the classroom* (pp. 137–160). Waxmann.

Korpershoek, H., Harms, T., de Boer, H., van Kuijk, M., & Doolaard, S. (2016). A meta-analysis of the effects of classroom management strategies and classroom management programs on students' academic, behavioral, emotional, and motivational outcomes. *Review of Educational Research, 86*(3), 643–680. https://doi.org/10.3102/0034654315626799

Kraft, M. A. (2020). Interpreting effect sizes of education interventions. *Educational Researcher, 49*(4), 241–253. https://doi.org/10.3102/0013189X20912798

Kulik, C.-L.C., & Kulik, J. A. (1982). Effects of ability grouping on secondary school students: A meta-analysis of evaluation findings. *American Educational Research Journal, 19*(3), 415–428. https://doi.org/10.3102/00028312019003415

Kultusministerkonferenz [KMK]. (2016). *Gemeinsame Initiative von Bund und Ländern zur Förderung leistungsstarker und potenziell besonders leistungsfähiger Schülerinnen und Schüler*. Beschluss der Kultusministerkonferenz vom 10.11.2016. https://www.kmk.org/themen/allgemeinbildende-schulen/individuelle-foerderung/foerderung-leistungsstaerkere.html

Kunter, M., Baumert, J., & Köller, O. (2007). Effective classroom management and the development of subject-related interest. *Learning and Instruction, 17*(5), 494–509. https://doi.org/10.1016/j.learninstruc.2007.09.002

Lichtenstein, S., & Slovic, P. (2006). The construction of preference: An overview. In S. Lichtenstein & P. Slovic (Eds.), *The construction of preference* (1st ed., pp. 1–40). Cambridge University Press.

Lüdtke, O., Trautwein, U., Kunter, M., & Baumert, J. (2007). Reliability and agreement of student ratings of the classroom environment: A reanalysis of TIMSS data. *Learning Environments Research, 9*(3), 215–230. https://doi.org/10.1007/s10984-006-9014-8

Marder, J., Thiel, F., & Göllner, R. (2023). Classroom management and students' mathematics achievement: The role of students' disruptive behavior and teacher classroom management. *Learning and Instruction*. https://doi.org/10.1016/j.learninstruc.2023.101746

Marsh, H. W. (1987). The big-fish-little-pond effect on academic self-concept. *Journal of Educational Psychology, 79*(3), 280–295. https://doi.org/10.1037/0022-0663.79.3.280

Marsh, H. W., Seaton, M., Trautwein, U., Lüdtke, O., Hau, K. T., O'Mara, A. J., & Craven, R. G. (2008). The big-fish–little-pond-effect stands up to critical scrutiny: Implications for theory, methodology, and future research. *Educational Psychology Review, 20*(3), 319–350. https://doi.org/10.1007/s10648-008-9075-6

McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics, 142*(2), 698–714. https://doi.org/10.1016/j.jeconom.2007.05.005

Möller, J. (2024). Ten years of dimensional comparison theory: On the development of a theory from educational psychology. *Educational Psychology Review, 36*(82), 544–560. https://doi.org/10.1007/s10648-024-09918-6

Neher-Asylbekov, S., & Wagner, I. (2023). Effects of out-of-school STEM learning environments on student interest: A critical systematic literature review. *Journal for STEM Education Research, 6*(1), 1–44. https://doi.org/10.1007/s41979-022-00080-8

OpenAI. (2024). *ChatGPT* (Version GPT-4) [Computer software]. https://chat.openai.com/

Organisation for Economic Co-operation and Development (OECD). (2016). *Teaching Excellence through Professional Learning and Policy Reform: Lessons from around the World*. Organisation for Economic Co-operation and Development. https://www.oecd-ilibrary.org/education/teaching-excellence-through-professional-learning-and-policy-reform_9789264252059-en

Pallas, A. M., Entwisle, D. R., Alexander, K. L., & Stluka, M. F. (1994). Ability-group effects: Instructional, social, or institutional? *Sociology of Education, 67*(1), 27–46. https://doi.org/10.2307/2112748

Patall, E. A. (2013). Constructing motivation through choice, interest, and interestingness. *Journal of Educational Psychology, 105*(2), 522–534. https://doi.org/10.1037/a0030307

Pianta, R. C., & Hamre, B. K. (2009). Conceptualization, measurement, and improvement of classroom processes: Standardized observation can leverage capacity. *Educational Researcher, 38*(2), 109–119. https://doi.org/10.3102/0013189X09332374

Praetorius, A.-K., & Gräsel, C. (2021). Noch immer auf der Suche nach dem heiligen Gral: Wie generisch oder fachspezifisch sind Dimensionen der Unterrichtsqualität? *Unterrichtswissenschaft, 49*(2), 167–188. https://doi.org/10.1007/s42010-021-00119-6

Praetorius, A.-K., Grünkorn, J., & Klieme, E. (2020a). Towards developing a theory of generic teaching quality: Origin, current status, and necessary next steps regarding the three basic dimensions model. *Zeitschrift Für Pädagogik Beiheft, 1*, 15–36. https://doi.org/10.3262/ZPB2001015

Praetorius, A.-K., Herrmann, C., Gerlach, E., Zülsdorf-Kersting, M., Heinitz, B., & Nehring, A. (2020b). Unterrichtsqualität in den Fachdidaktiken im deutschsprachigen Raum – zwischen Generik und Fachspezifik. *Unterrichtswissenschaft, 48*(3), 409–446. https://doi.org/10.1007/s42010-020-00082-8

Praetorius, A.-K., Klieme, E., Herbert, B., & Pinger, P. (2018). Generic dimensions of teaching quality: The German framework of three basic dimensions. *ZDM - Mathematics Education, 50*(3), 407–426. https://doi.org/10.1007/s11858-018-0918-4

Preckel, F., Golle, J., Grabner, R., Jarvin, L., Kozbelt, A., Müllensiefen, D., Olszewski-Kubilius, P., Schneider, W., Subotnik, R., Vock, M., & Worrell, F. C. (2020). Talent development in achievement domains: A psychological framework for within- and cross-domain research. *Perspectives on Psychological Science, 15*(3), 691–722. https://doi.org/10.1177/1745691619895030

R Core Team. (2024). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Reis, S. M., & Renzulli, J. S. (2004). Current research on the social and emotional development of gifted and talented students: Good news and future possibilities. *Psychology in the Schools, 41*(1), 119–130. https://doi.org/10.1002/pits.10144

Reis, S. M., & Renzulli, J. S. (2023). The schoolwide enrichment model: A focus on student strengths & interests. In E. J. Gubbins, K. S. McMillen, J. S. Renzulli, R. D. Eckert, & C. A. Little (Eds.), *Systems and models for developing programs for the gifted and talented* (2nd ed., pp. 323–352). Routledge.

Research Data Center for Education (FDZ Bildung). (2023). Data access to educational research data through the FDZ Bildung. Leibniz Institute for Research and Information in Education (DIPF). https://www.fdz-bildung.de/home

Richards, M., & Omdal, S. (2007). Effects of tiered instruction on academic performance in a secondary science course. *Journal of Advanced Academics, 18*, 424–453. https://doi.org/10.4219/jaa-2007-499

Röhl, S., Davison, K. T., & Bardach, L. (2024). *Adaptivity Explains Heterogeneity in Students' Perceptions.* Preprint. https://doi.org/10.31219/osf.io/mbntz

Rosseel, Y., Jorgensen, T. D., Wilde, L. D., Oberski, D., Byrnes, J., Vanbrabant, L., Savalei, V., Merkle, E., Hallquist, M., Rhemtulla, M., Katsikatsou, M., Barendse, M., Rockwood, N., Scharf, F., Du, H., Jamil, H., & Classe, F. (2024). *lavaan: Latent Variable Analysis* (Version 0.6–19) [Computer software]. https://cran.r-project.org/web/packages/lavaan/index.html

Sahin, A. (2013). STEM project-based learning. In R. M. Capraro, M. M. Capraro, & J. R. Morgan (Eds.), *STEM project-based learning: An integrated science, technology, engineering, and mathematics (STEM) approach* (pp. 59–64). SensePublishers.

Senden, B., Nilsen, T., & Blömeke, S. (2022). Instructional quality: A review of conceptualizations, measurement approaches, and research findings. In M. Blikstad-Balas, K. Klette, & M. Tengberg (Eds.), *Ways of analyzing teaching quality* (pp. 140–172). Scandinavian University Press.

Steenbergen-Hu, S., Makel, M. C., & Olszewski-Kubilius, P. (2016). What one hundred years of research says about the effects of ability grouping and acceleration on K–12 students' academic achievement: Findings of two second-order meta-analyses. *Review of Educational Research, 86*(4), 849–899. https://doi.org/10.3102/0034654316675417

Suprayogi, M. N., Valcke, M., & Godwin, R. (2017). Teachers and their implementation of differentiated instruction in the classroom. *Teaching and Teacher Education, 67*, 291–301. https://doi.org/10.1016/j.tate.2017.06.020

Tajfel, H., & Turner, J. C. (1979). An integrative theory of intergroup conflict. In W. G. Austin & S. Worchel (Eds.), *The Social Psychology of Intergroup Relations* (pp. 33–47). Brooks/Cole.

Taulien, M. (2001). *Mathematiktest zum MINT-Projekt. Eingangs- und Retest für Hectorianer und Kontrollgruppenschüler.* Geschäftsstelle des Hector-Seminars.

Teig, N., Scherer, R., & Nilsen, T. (2019). I know I can, but do I have the time? The role of teachers' self-efficacy and perceived time constraints in implementing cognitive-activation strategies in science. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2019.01697

Tieso, C. (2003). Ability grouping is not just tracking anymore. *Roeper Review, 26*(1), 29–36. https://doi.org/10.1080/02783190309554236

Tomlinson, C. A., Brighton, C., Hertberg, H., Callahan, C. M., Moon, T. R., Brimijoin, K., Conover, L. A., & Reynolds, T. (2003). Differentiating instruction in response to student readiness, interest, and learning profile in academically diverse classrooms: A review of literature. *Journal for the Education of the Gifted, 27*(2–3), 119–145. https://doi.org/10.1177/016235320302700203

Trautwein, U., Golle, J., Jaggy, A.-K., Hasselhorn, M., & Nagengast, B. (2023). Mutual benefits for research and practice: Randomized controlled trials in the Hector Children's Academy Program. *Annals of the New York Academy of Sciences, 1530*(1), 96–104. https://doi.org/10.1111/nyas.15074

Wagner, W., Göllner, R., Werth, S., Voss, T., Schmitz, B., & Trautwein, U. (2016). Student and teacher ratings of instructional quality: Consistency of ratings over time, agreement, and predictive power. *Journal of Educational Psychology, 108*(5), 705–721. https://doi.org/10.1037/edu0000075

Weinert, F. E., & Helmke, A. (1995). Interclassroom differences in instructional quality and interindividual differences in cognitive development. *Educational Psychologist, 30*(1), 15–20. https://doi.org/10.1207/s15326985ep3001_2

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., & Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.